SSM

Scuola Superiore Meridionale

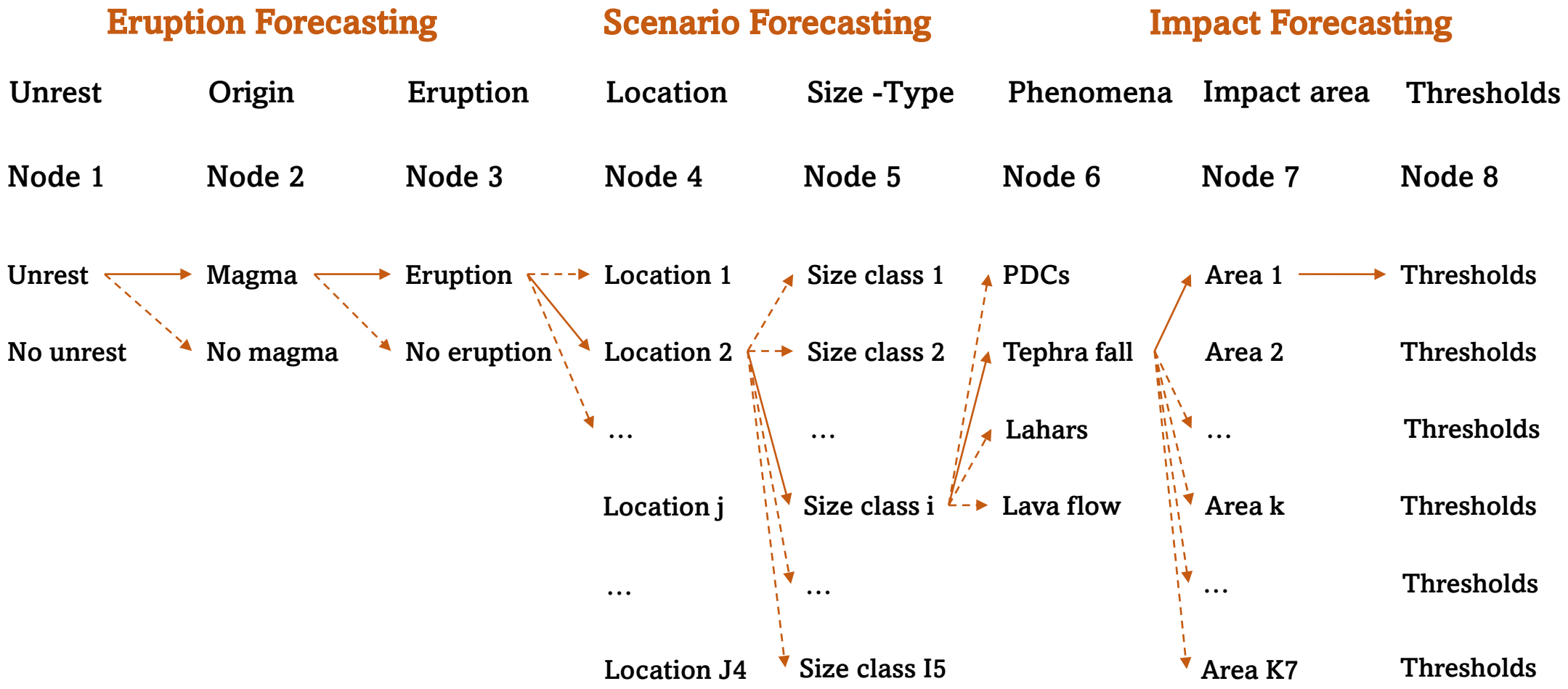
pyBET-evo - tutorial

Salvatore Ferrara

salvatore.ferrara-ssm@unina.it

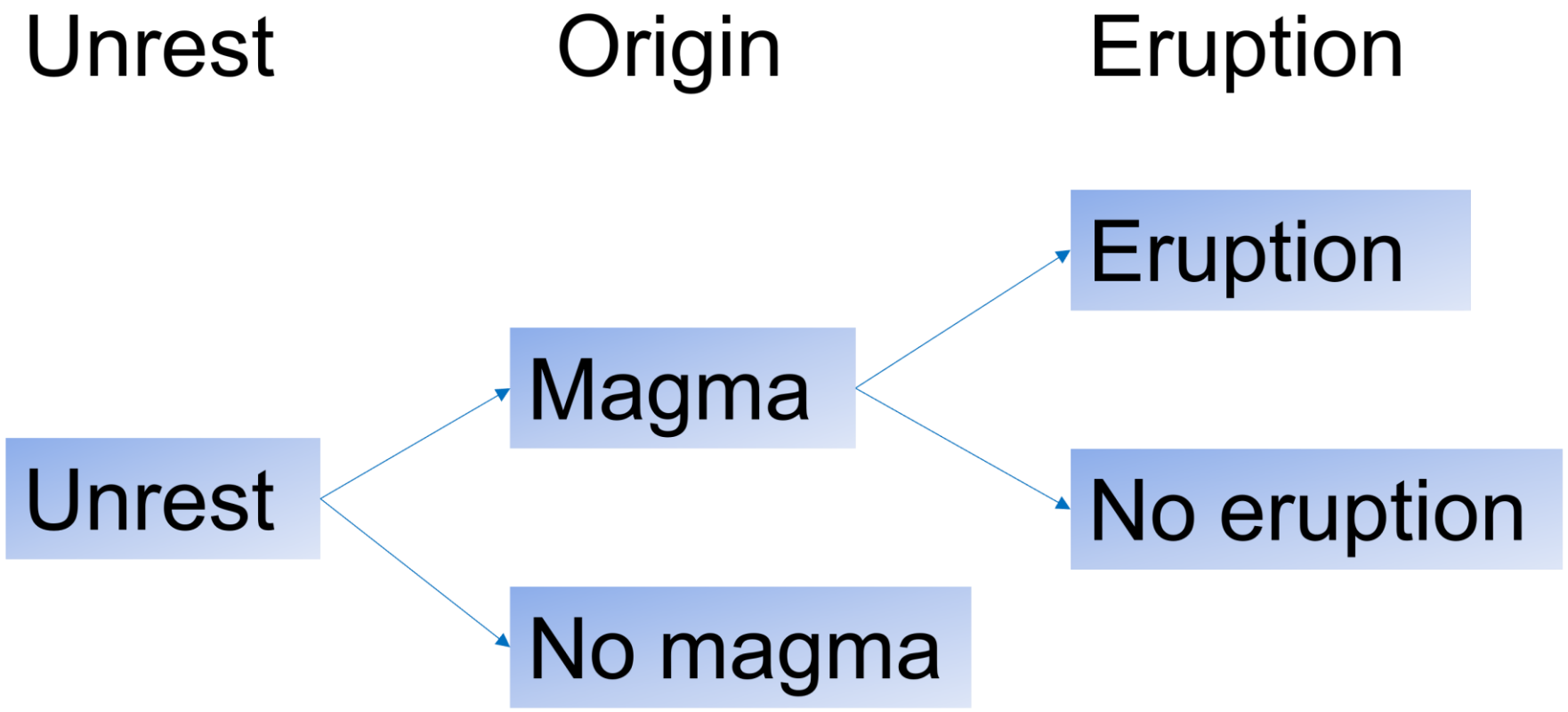
Part 1: theory

The Bayesian Event Tree (BET)



The Bayesian Event Tree (BET) is a tree model that decomposes the calculation of the probability for each specific volcanic event of interest into a chain of conditional probabilities (Marzocchi et al., 2004, 2008). Within the event tree, individual probabilities are represented as random variables. This approach is employed to consider both aleatory and epistemic uncertainties, which are related to the mean and the variance of the corresponding distribution.

Eruption Forecasting



The goal of pyBET-evo is to provide the user with a software capable of evaluating the probability that the mechanism underlying the unrest of a volcano is a shallow magma movement and the probability of eruption as a function of monitoring anomalies at different time-steps. The code therefore allows to perform with a simple click operations that would otherwise have taken a long time, depending on the number of time-steps.

Eruption Forecasting

Nodes 2 and 3 of BET (magmatic unrest and eruption) represent mutually exclusive events and are defined by Beta distributions:

$$\Phi \sim Beta(\alpha, \beta)$$

where Φ is the random variable that represents the probability at the node under consideration.

The Beta distribution is frequently selected due to its definition within the range [0, 1] and its status as the conjugate prior distribution in the binomial model (Gelman et al., 1995). This characteristic facilitates the integration of expert opinions or model output with observational data.

The parameters α and β are obtained by inverting the following equations:

$$E[\Phi] = \frac{\alpha}{\alpha + \beta}$$

$$\sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

where $E[\Phi]$ is the expected value of the Beta distribution, i.e., the mean, and σ^2 is the variance.

Managing the monitoring information through fuzzy logic

The BET approach employs fuzzy logic (Zadeh, 1965), wherein each monitoring measurement is assigned a degree of anomaly (z). Thus, by defining two threshold values, a monitoring observation may not be entirely non-anomalous ($z = 0$) or completely anomalous ($z = 1$) but can exhibit anomaly to a certain degree ($0 < z < 1$).

When an anomaly is characterized by high values of the monitoring variable:

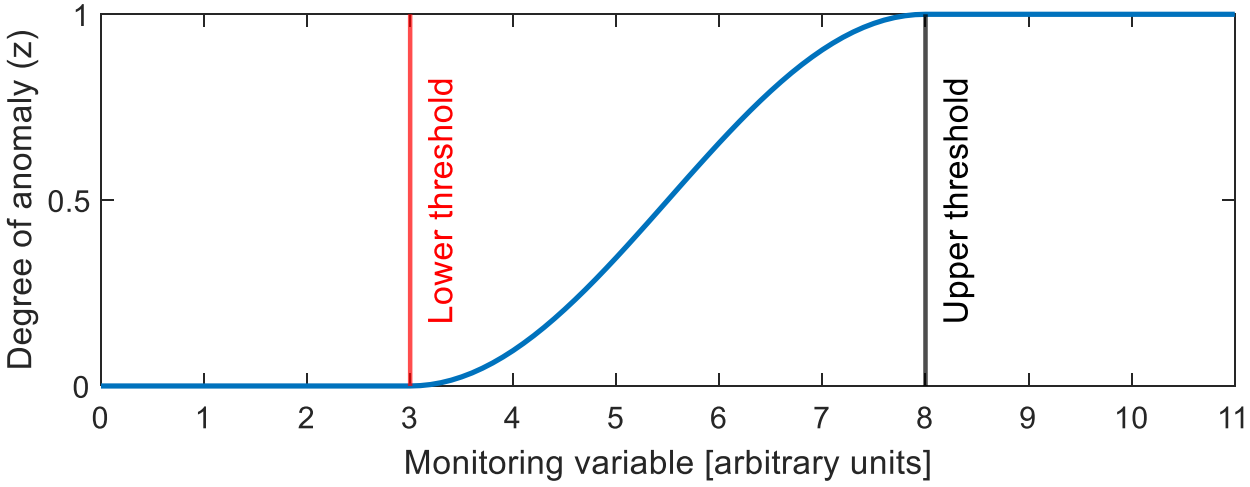
$$z_i = \begin{cases} 1 \text{ if } & a_i \geq a_{max} \\ \frac{1}{2}\left\{sin\left[\pi \frac{(a_i - a_{min})}{(a_{max} - a_{min})} - \frac{\pi}{2}\right] 1\right\} & \text{if } a_{min} < a_i < a_{max} \\ 0 \text{ if } & a_i \leq a_{min} \end{cases}$$

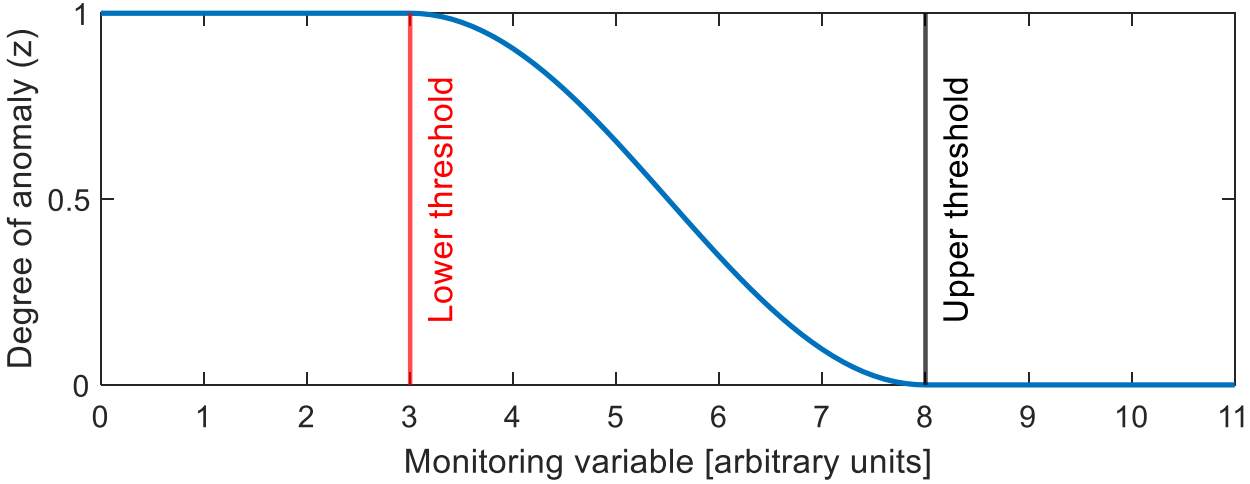
When it is characterized by low values:

$$z_i = \begin{cases} 0 \text{ if } & a_i \geq a_{max} \\ \frac{1}{2}\left\{sin\left[\pi \frac{(a_i - a_{min})}{(a_{max} - a_{min})} + \frac{\pi}{2}\right] + 1\right\} & \text{if } a_{min} < a_i < a_{max} \\ 1 \text{ if } & a_i \leq a_{min} \end{cases}$$

where a_{min} and a_{max} are, respectively, the minimum and maximum value of the monitoring variable a which defines when the parameter is certainly anomalous or not.

Managing the monitoring information through fuzzy logic





Variation of the degree of anomaly z as a function of the monitoring variable a.
Marzocchi et al. (2024).

The monitoring information is then summarized by the cumulative degree of anomaly (hereafter anomaly score):

$$Z = \sum_{i=1}^{N} z_i \omega_i$$

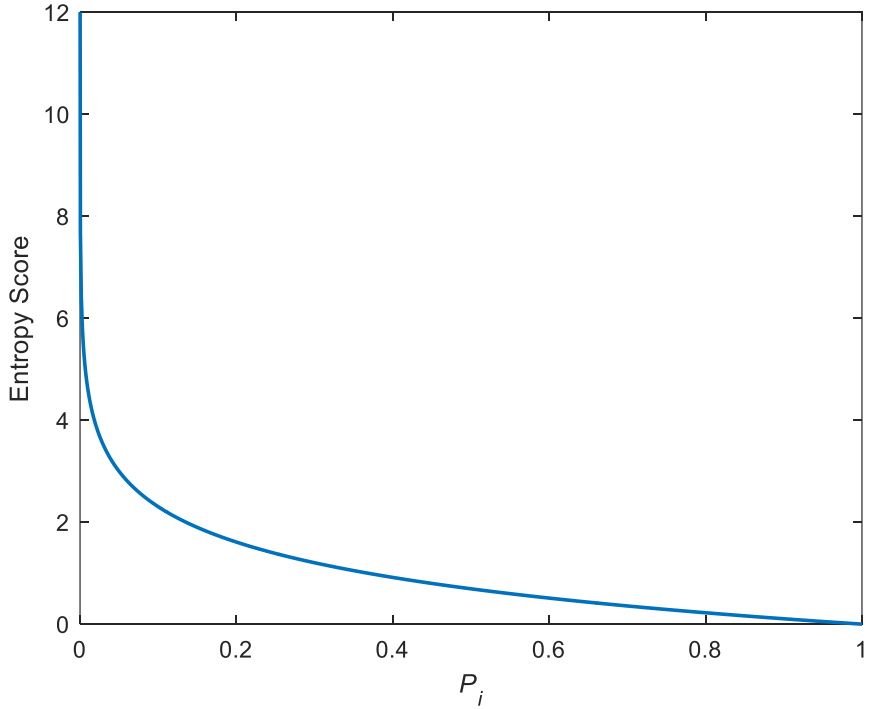
where ω_i is the weight of the *i*-th anomaly.
Weights and thresholds are usually the result of an experts' elicitation experiment (Selva et al., 2012; Ferrara et al., 2025)

From monitoring measures to probabilities through an entropy-based approach

The concept of entropy is used in various fields, such as physics, thermodynamics, and information theory, and it generally refers to a measure of disorder, randomness, and uncertainty (Marzocchi et al., 2024).

If the event we are interested in is a volcanic eruption, we have two outcomes that we name "event" (E) and "non event" $(\bar{E})$, each one characterized by a probability $P_i = P_E$ and $P_i = 1 - P_E$, respectively.

The term $-\ln(P_i)$ is called entropy score (Daley and Vere-Jones, 2005) and it is particularly important being a measure of the unpredictability of the particular outcome *i*, i.e., $-\ln(P_i)$ is a measure of the unpredictability of E : when E is perfectly predictable, the entropy score $-\ln(P_i)$ is 0, whereas when E has a very low probability the entropy score goes towards infinity.



Relationship between P_i and the entropy score.

From monitoring measures to probabilities through an entropy-based approach

We can define the unpredictability of $\bar{E}$, $-\ln(1 - P_E)$, as the predictability of E, which goes towards infinity when P_E is approaching to 1, and it goes towards to 0 when E becomes more unlikely.

The simplest way in which we may use the monitoring information to linearly modify the predictability of E as a function of the anomaly score is:

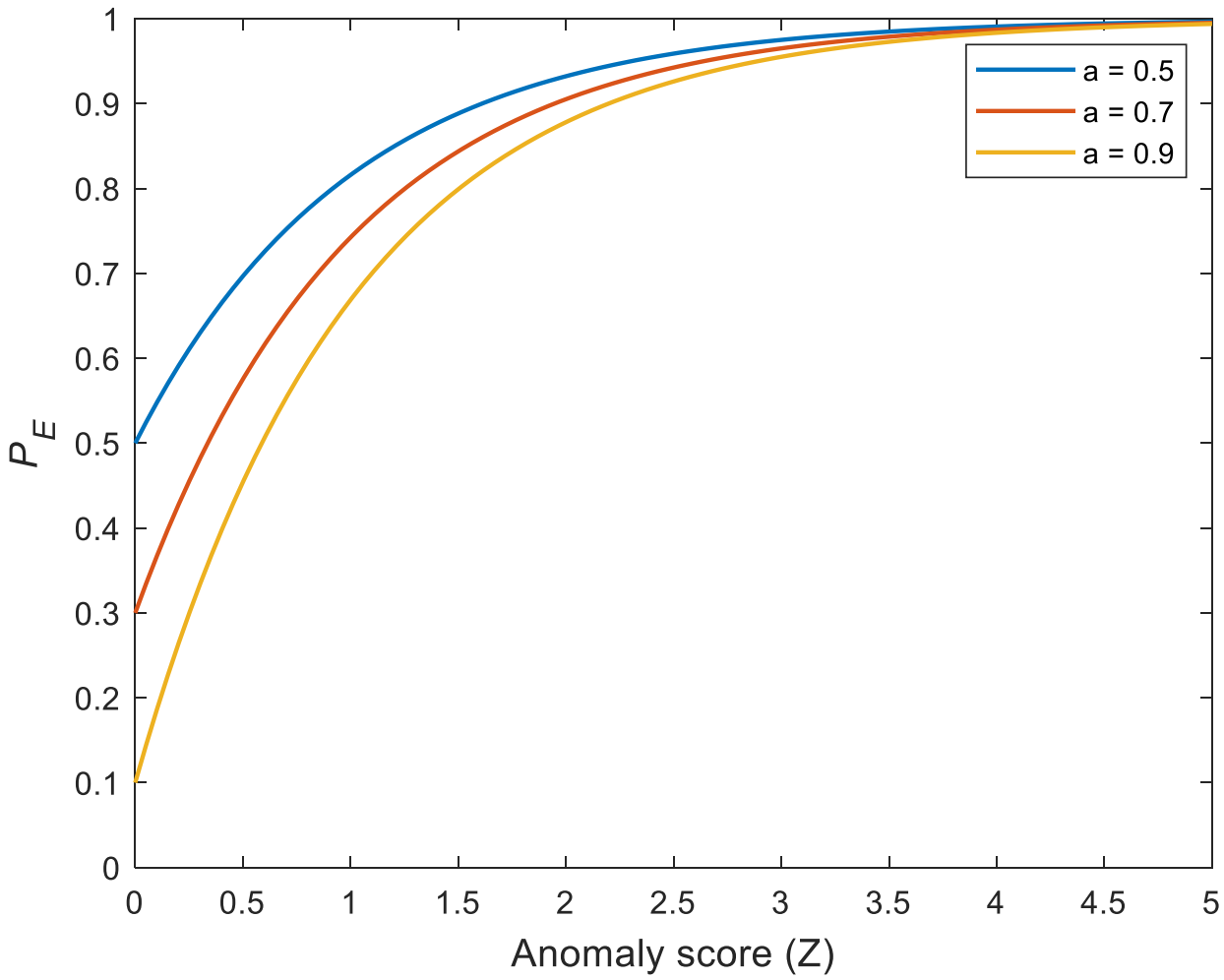
$$-\ln(1 - P_E) = k + Z$$

The parameter *k* sets a lower boundary of predictability when the monitoring system of the volcano does not detect any anomaly ($Z = 0$), accounting for any potential anomaly that the current monitoring system is not able to detect.

We can establish how P_E evolves as a function of the anomalies detected:

$$P_E = 1 - e^{-(k+Z)} = 1 - ae^{-Z}$$

From monitoring measures to probabilities through an entropy-based approach



Relationship between Z and P_E for three different values of a.

$$P_E = 1 - e^{-(k+Z)} = 1 - ae^{-Z}$$

This relationship has some important features that are worth being remarked:

- The relationship between Z and P_E is monotonically increasing, so that the higher the value of Z, the more likely is the occurrence of E.
- Such kind of relationship implies that the largest increase in P_E occurs when one of the monitoring variables shows some degree of anomaly. As more monitored variables become anomalous, P_E keeps on rising, but slower.
- It states that P_E depends only on the anomaly score Z.

From monitoring measures to probabilities through an entropy-based approach

In our framework we consider P_E as the expected value of a Beta distribution. We can express the confidence on the forecast in terms of "equivalent number of data" Λ, which is defined as:

$$\Lambda = \alpha + \beta - 1$$

The term "equivalent number of data" comes from the fact that $\alpha + \beta$ depends on the number of observations used to set up the distribution. The minimum value of Λ is 1, given that in the most uninformative case $\alpha = \beta = 1$ (when the expected value is equal to 0.5), which corresponds to a uniform distribution, and it increases if we consider more data.

Now, we can easily obtain the parameters of the Beta distribution by solving this simple system of equations:

$$\begin{cases} E[\Phi] = \dfrac{\alpha}{\alpha + \beta} \\ \Lambda = \alpha + \beta - 1 \end{cases}$$

Rescaling the probability of eruption for shorter time-windows

In short-term application BET usually refers to time windows of $\tau = 1$ month, but we may need forecasts on time windows of varying length (Δt).
Following Selva et al. (2014) we can rescale the probability of eruption for shorter time windows:

$$[\phi_3(t_0, t_0 + \Delta t)] = [\phi_3(t_0, t_0 + \tau)]F(t_0 + \Delta t)$$

where $F(t)$ is the CDF of the following PDF:

$$f(t) = ke^{\frac{-t}{T_c}}$$

Where T_c is the characteristic time which is set to 1 week and k is a normalizing factor such that $\int_{t_0}^{\tau} f(t)dt = 1$.

What happens if there's no unrest?

If the volcano is in a quiet state we can calculate the probability of unrest, the probability of magmatic unrest and the probability of eruption in a given time window using data relating to past frequencies and prior information.

To do so, BET uses Bayesian inference (Gelman, 1965):

$$[\phi|x] \propto [\phi_{prior}][x|\phi]$$

where $[\phi_{prior}]$ is the prior distribution and $[x|\phi]$ is the likelihood function. In the BET approach, the prior is modelled through a Beta distribution, while the likelihood is modelled through a binomial distribution. The choice of the Beta and Binomial distributions is not random, since they are conjugate distributions, i.e., the posterior distribution can be easily obtained as:

$$\Phi|n, x \sim Beta(\alpha + x, \beta + n - x)$$

Where n is the number of non-overlapping time windows, x is the number of successes (i.e., the total number of windows that include the event we want to forecast), and α and β are obtained by assigning mean and variance through experts' judgement.

For long-term forecasts we can rescale the probabilities by assuming a uniform behaviour inside of the time-window τ:

$$[\phi_{1,2,3}(t_0, t_0 + \Delta t)] = [\phi_{1,2,3}(t_0, t_0 + \tau)]\frac{\Delta t}{\tau}$$

Part 2: how to use the code

Main Settings

```
Unrest = False #True if you want to compute probabilities from monitoring data, False if you want to compute probabilities from long-term data
n_sample = 100000
dt = 1 #lower or equal than one
prc = [10, 50, 90]
```

- Unrest: True or False depending on if we want to make a short- or long-term forecast.
- n_sample: the size of the samples of the PDFs, larger samples are less noisy, but are more computationally intensive.
- dt: the fraction of the original time window in which we want to make the forecast.
- prc: the percentiles we want as output, other than the mean.

```
if Unrest == False: #fill if you want to compute the long-term probability

    #Long-term Unrest
    n1 = 306
    x1 = 7
    mean_prior_1 = 0.5
    lambda_prior_1 = 1

    #Long-term Magmatic Unrest
    n2 = 7
    x2 = 1
    mean_prior_2 = 0.5
    lambda_prior_2 = 1

    #Long-term Eruption
    n3 = 3.7
    x3 = 0
    mean_prior_3 = 0.5
    lambda_prior_3 = 1
```

Long-term settings

If we want to evaluate the long-term probability we need to provide the following parameters;

- n: number of time windows
- x: number of successes
- mean_prior: average of the prior distribution
- lambda_prior: equivalent number of data of the prior distribution

Main Settings

```
Unrest = True #True if you want to compute probabilities from monitoring data, False if you want to compute probabilities from long-term data
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- dt: the fraction of the original time window in which we want to make the forecast.
- prc: the percentiles we want as output, other than the mean.

```
if Unrest == True: #fill if you want to compute the short-term probability

    #Anomaly Function Parameters
    a = 0.9
    lambda_short_term = 1

    #Monitoring file (path or file name if it's in the same directory of the code)
    Monitoring = 'Monitoring.xlsx'
```

Short-term settings

If we want to evaluate the short-term probability we need to provide the following parameters;

- a: e^{-k} (see slide 10)
- lambda_short_term: equivalent number of data
- Monitoring: name of the monitoring excel file that we must prepare.

Monitoring file: Magmatic Unrest

	A	B	C	D	E
1	Cumulative Uplift	Daily number of earthquakes	Fractures		
2	-0.3	0	0		
3	1	0	0		
4	0.8	0	0		
5	0.8	0	0		
6	0.3	0	0		
7	-0.2	0	0		
8	1.3	0	0		
9	4.4	0	0		
10	8.9	0	0		
11	12.5	0	0		
12	13.6	0	0		
13	14.2	0	0		
14	14.7	1	0		
15	15	0	0		
16	16.1	0	0		
17	15.5	0	0		
18	15.1	0	0		
19	13.1	0	1		
20	14.8	0	1		
21	17.1	1	1		
22	21.9	6	1		
23	23.6	1	0		
24	23.2	1	0		
25	22	3	0		
26	21.3	2	0		
27	24.9	2	0		

Magmatic Unrest | Magmatic Unrest - Parameters | Eruption | Eruption - Parameters

Each colum represents a different parameter. Each row represents a different time-step.

Monitoring file: Magmatic Unrest - Parameters

	A	B	C	D	E	F	G
1		Cumulative Uplift	Daily number of earthquakes	Fractures			
2	Weight	1	0.9	0.2			
3	T1	5	2	0			
4	T2	15	20	1			

Magmatic Unrest | Magmatic Unrest - Parameters | Eruption | Eruption - Parameters

Each colum represents the same parameters reported in the Magmatic Unrest sheet. We must assign weights and thresholds to each of these parameters. As we mentioned in the theoretical section, BET uses a fuzzy logic. T1 and T2 are the anomaly thresholds, if the parameter is considered anomalous for high values we set T1 > T2, if the parameter is anomalous for low values, we set T2 > T1. If the parameter is boolean, i.e., its simple observation results in an anomaly, we set T1 = 0 and T2 = 1.

Monitoring file: Eruption

Eruption and Eruption - Parameters should be set the same way

Output: long-term

	A	B	C	D	E
1		**Mean**	**10-th Percentile**	**50-th Percentile**	**90-th Percentile**
2	**Probability of Unrest**	0.025936026	0.015211781	0.024942078	0.038004703
3	**Probability of Magmatic Unrest**	0.005761225	0.001513348	0.004802268	0.011270045
4	**Probability of Eruption**	0.001010519	6.96311E-05	0.000571488	0.002477785

The output of the long-term forecast is an excel file that contains mean and percentiles of the long-term absolute probability distributions of unrest, magmatic unrest and eruption.

Output: short-term

	A	B	C	D	E	F	G	H
1	Mean Magmatic Unrest	10-th Percentile Magmatic Unrest	50-th Percentile Magmatic Unrest	90-th Percentile Magmatic Unrest	Mean Eruption	10-th percentile Eruption	50-th percentile Eruption	90-th percentile Eruption
2	0.099848819	4.80649E-06	0.014565906	0.341105419	0.010127533	8.22439E-10	5.49696E-05	0.020345289
3	0.10021211	4.56294E-06	0.014405719	0.343981952	0.010068921	8.71473E-10	5.22077E-05	0.019474758
4	0.100593373	4.48015E-06	0.014094584	0.347835339	0.010032686	8.31524E-10	5.13974E-05	0.019691547
5	0.100452789	4.65239E-06	0.014629748	0.343969783	0.010042334	7.46501E-10	5.12835E-05	0.01990752
6	0.100197941	4.44925E-06	0.014742273	0.343400605	0.009973745	7.99411E-10	5.31062E-05	0.019374258
7	0.10027741	4.44625E-06	0.014747699	0.343964381	0.010072852	7.46276E-10	5.16241E-05	0.019981142
8	0.100685943	4.51472E-06	0.014632796	0.346606936	0.009986718	7.02544E-10	5.08554E-05	0.01952695
9	0.099150899	4.34418E-06	0.014037882	0.341734216	0.009797575	7.22705E-10	5.10012E-05	0.019242669
10	0.352467573	0.028905219	0.296645136	0.776142567	0.035489152	6.64543E-07	0.002519034	0.108060408
11	0.615977289	0.191932513	0.659351715	0.96054233	0.061780106	1.93503E-06	0.006825154	0.202081007
12	0.652771339	0.229834471	0.709169482	0.972948034	0.064998834	2.51863E-06	0.007595572	0.215678679
13	0.664112027	0.24196037	0.726122432	0.976039607	0.066414401	2.49719E-06	0.007953028	0.21866422
14	0.667086884	0.244928978	0.729276995	0.976989462	0.066767131	2.59799E-06	0.008004491	0.221194077
15	0.670079259	0.24614958	0.734094661	0.978179306	0.065991733	2.48611E-06	0.007791665	0.217190663
16	0.668194937	0.246663785	0.730746507	0.97767163	0.066740586	2.47815E-06	0.007960289	0.222201498
17	0.668669454	0.247622675	0.730186358	0.977310959	0.065999066	2.65383E-06	0.007924508	0.218796811
18	0.669393339	0.24808223	0.73259768	0.977882203	0.066878009	2.53076E-06	0.007996076	0.220030349
19	0.703455437	0.287140256	0.777348621	0.986694407	0.277688272	0.022703152	0.214691838	0.641937697
20	0.728929682	0.322939699	0.80990342	0.990844359	0.288680259	0.024647544	0.227109281	0.660637335
21	0.727278939	0.320151821	0.807716234	0.99057742	0.288798042	0.024825122	0.228112341	0.658924826
22	0.755138863	0.357459407	0.843731627	0.994422702	0.299132066	0.026622852	0.238185032	0.677334713
23	0.668043487	0.246326515	0.730551036	0.977584864	0.066112584	2.73667E-06	0.007844917	0.2186495
24	0.668781938	0.247583347	0.731838597	0.977563717	0.066906754	2.53963E-06	0.007988265	0.221921228
25	0.672286767	0.249398238	0.736158902	0.978970548	0.067648916	2.53882E-06	0.007978985	0.223993266
26	0.667580458	0.246273752	0.729080889	0.977327618	0.066299998	2.42767E-06	0.007880739	0.217213991
27	0.669818051	0.248005251	0.733800258	0.977702477	0.067286008	2.6488E-06	0.008158526	0.22293159
28	0.672456593	0.24956079	0.736965722	0.978818421	0.067224944	2.47536E-06	0.007934804	0.221577511
29	0.718295766	0.307294988	0.796800379	0.988798348	0.071659523	2.74561E-06	0.008982304	0.241475317
30	0.668005903	0.246186212	0.730017327	0.977126388	0.066964806	2.63786E-06	0.008087541	0.220640731
31	0.668340659	0.244741503	0.73113111	0.977445232	0.067182921	2.61961E-06	0.008100099	0.22246137

The output of the short-term forecast is an excel file that contains the variation of mean and percentiles of the short-term absolute probability distributions of magmatic unrest and eruption.

References

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